

Modeling, Simulation and Design of a Ge-on-Si PIN Photodetector IEEE 802.3bs DR4 Link Integration

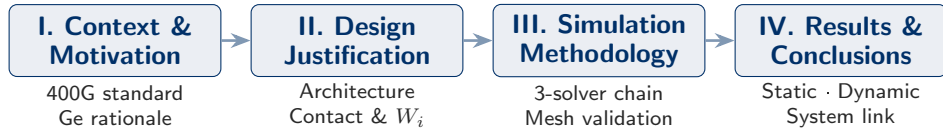
Design of a 400 Gb/s Si-Photonics Transceiver for IEEE 802.3 DR4 Intra-DC
Interconnects

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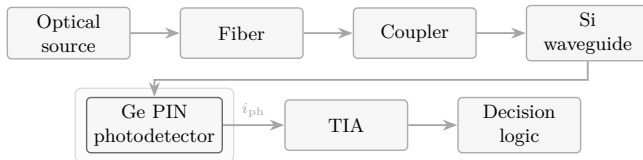
Presentation Outline



The 400 Gb/s DR4 Standard and Intra-Data-Centre Bottlenecks

Driving Standard

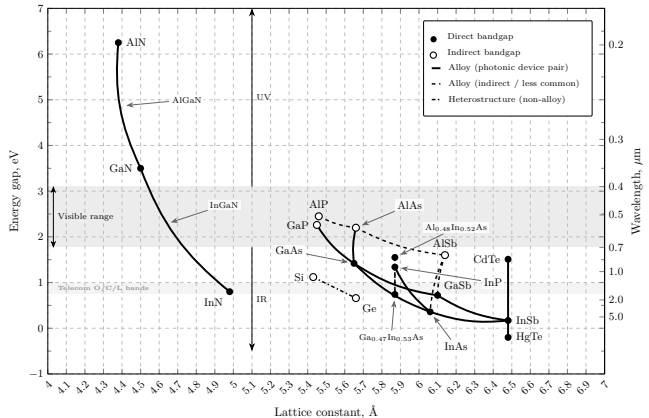
- ▶ Data-centre traffic grows at $> 30\%/yr$; inter-rack links are the primary bottleneck
- ▶ **IEEE 802.3bs DR4:** 4×106.25 Gb/s PAM-4 over 500 m SMF at $\lambda = 1310$ nm
- ▶ Per-lane requirement: $f_{3dB} > 39.8$ GHz; sensitivity < -9 dBm (KP4 FEC)
- ▶ **Receiver constraint:** photodetector limits both speed and noise floor



Material Selection: Germanium on Silicon Integration

Four Decisive Criteria

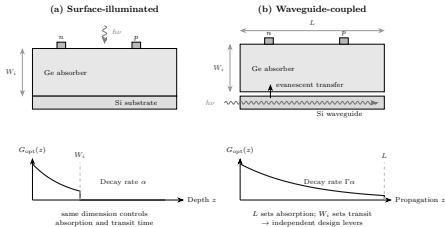
- **CMOS Compatibility:** Monolithic Si-photonics integration without expensive III-V bonding
- **O-Band Absorption:** $\alpha \approx 3 \times 10^3 \text{ cm}^{-1}$ at 1310 nm; direct $E_g = 0.80 \text{ eV}$
- **High Mobility:** $\mu_e = 3900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ ($3 \times$ faster than Si)
- **Mature Epitaxy:** Graded buffer limits dislocation density $< 10^7 \text{ cm}^{-2}$



Device Architecture: Waveguide PIN Configuration

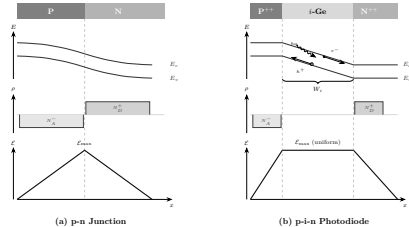
Waveguide-Coupled Advantage

- **Decoupled Path:** Waveguide geometry separates transit length (W_i) from absorption length (L)
- **Co-optimization:** Choose W_i for speed and L for responsivity independently without compromise



PIN Junction Electrostatics

- **Uniform Field:** Potentials drop linearly, yielding carrier velocity saturation throughout the intrinsic region
- **Low Capacitance:** Depletion width W_i directly lowers junction capacitance $C_j = \epsilon A / W_i$



Design Choices, Trade-offs & Simulation Chain

Design Levers

U-contact: Reduces R_a by 36–40%

Mesa: $W_m=5\text{ }\mu\text{m}$ balances C_j and series resistance

Thickness: $W_i=350\text{ nm}$ balances speed vs. C_j

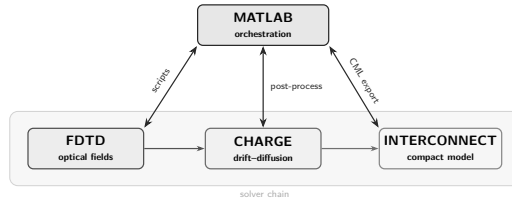
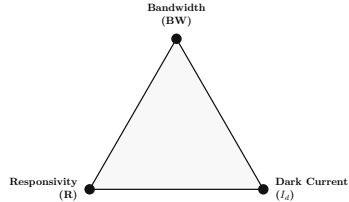
Triangle: Speed – Responsivity – Noise

Solver Chain

FDTD $\rightarrow$ 3D $G_{\text{opt}}(\mathbf{r})$

CHARGE $\rightarrow I_d, R, C_j, S21$

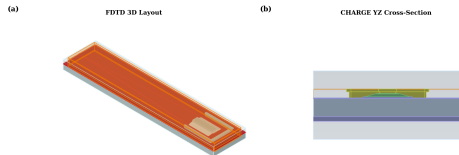
INTERCONNECT $\rightarrow$ PAM-4 eye, BER



Device Layout: Solver Setup & GDS Mask

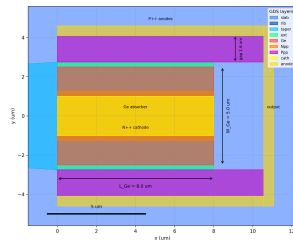
Solver Mesh & Layout

- **FDTD & DEVICE:** 3D evanescent waveguide taper and 2D triangular refinement mesh
- **Convergence:** Refinement < 20 nm at Ge/Si interfaces ensures numerical stability



GDS Mask & DRC/LVS Checks

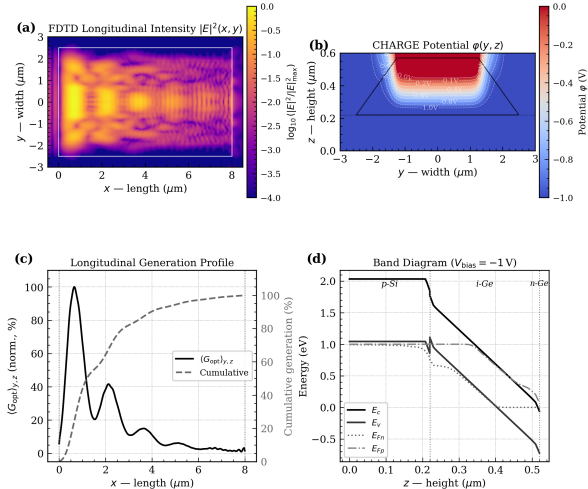
- **Scripted Layout:** KLayout cell with 10 logical process layers (Si slab, Ge, implants, contacts)
- **Validation:** Automated checks confirm mask matches solver design variables (W_i , L , W_m)



Solver Validation: Field Profiles at -1.0 V

What the Profiles Confirm

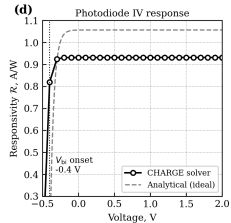
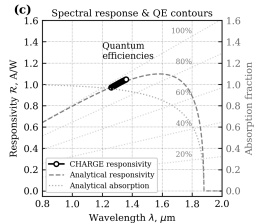
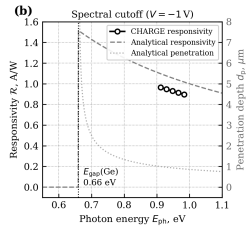
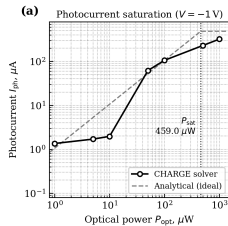
- ▶ (a) **FDTD intensity** $|E|^2(x, y)$: evanescent coupling into Ge absorber confirms mode overlap
- ▶ (b) **Electrostatic potential** $\varphi(y, z)$: uniform drop across intrinsic Ge confirms full depletion at -1.0 V
- ▶ (c) **Optical generation** $G_{\text{opt}}(x)$: exponential decay + cumulative % validates Beer-Lambert model
- ▶ (d) **Band diagram** E_c, E_v, E_{fn}, E_{fp} vs. z : confirms complete electron-hole separation



Result 1: Responsivity and Quantum Efficiency

Spectral Efficiency

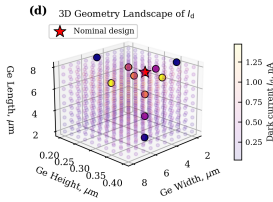
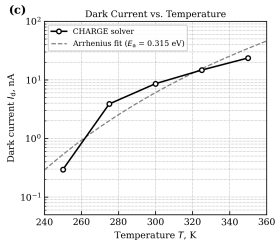
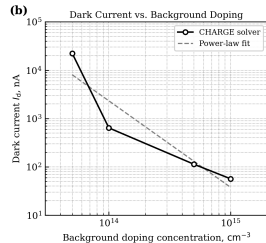
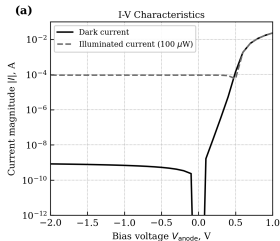
- **FDTD (optical limit):** $\mathcal{R} = 0.968 \text{ A/W}$, EQE= 91.63 %
- **CHARGE (transport):** $\mathcal{R} = 0.931 \text{ A/W}$, EQE= 88.08 %, IQE= 96.23 %
- 3.55 % EQE gap = optical coupling loss; 96 % IQE = near-perfect carrier collection
- **Benchmark** (Shi et al., 2024): $\mathcal{R} = 0.95 \text{ A/W}$ (within 2 %)



Result 2: Dark Current and Thermal Activation

Noise Floor Characterisation

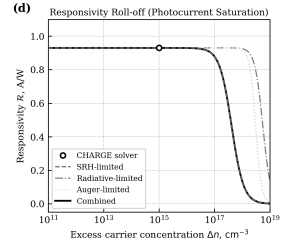
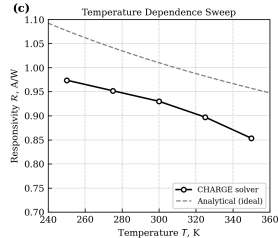
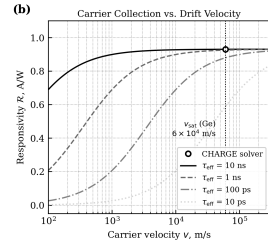
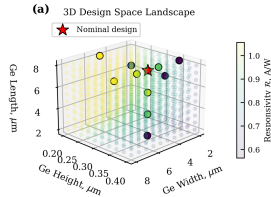
- ▶ $I_d = 0.684 \text{ nA}$ at -1.0 V ($1.9\times$ lower than the Shi et al. (2024) benchmark of 1.3 nA)
- ▶ Saturation confirms SRH bulk generation (not surface leakage or tunnelling)
- ▶ Doubles every $\Delta T \approx 10 \text{ K}$; activation energy $E_a \approx 0.315 \text{ eV}$



Result 3: Design Sweeps and Responsivity Limits

Responsivity Sweeps

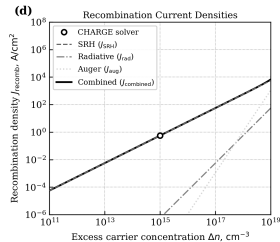
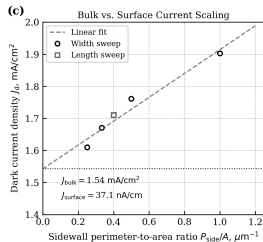
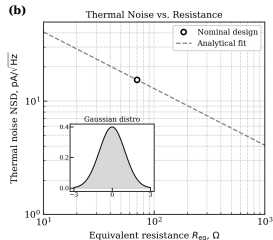
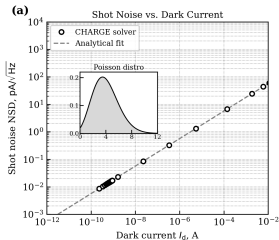
- **3D Landscape:** Responsivity mapped vs. Ge mesa width and length
- **Collection:** Approaching unity as drift velocity exceeds saturation threshold
- **Recombination:** Roll-off at high excess carrier concentration ($> 10^{18} \text{ cm}^{-3}$)



Result 4: Noise Performance and Recombination Limits

Noise and Recombination Limits

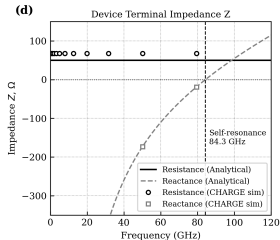
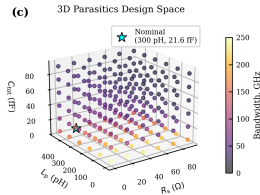
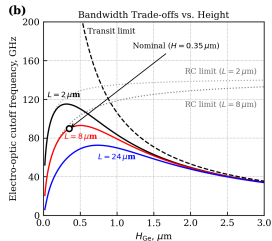
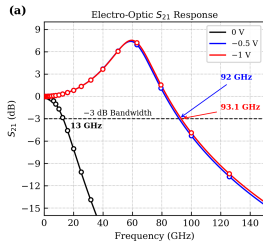
- ▶ **Shot Noise:** $\sqrt{2qI_d} \approx 14.8 \text{ fA}/\sqrt{\text{Hz}}$ matches analytical limit at -1.0 V nominal bias
- ▶ **Thermal Noise:** $\sqrt{4kT/R} \approx 15.3 \text{ pA}/\sqrt{\text{Hz}}$ at nominal total loop resistance of 70Ω
- ▶ **Perimeter Scaling:** Dark current scales with perimeter-to-area ratio, highlighting surface leakage



Result 5: Small-Signal AC Bandwidth

Frequency Domain Performance

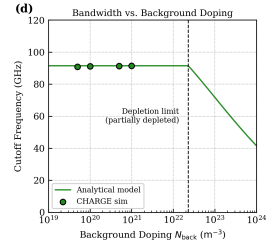
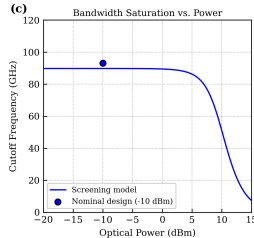
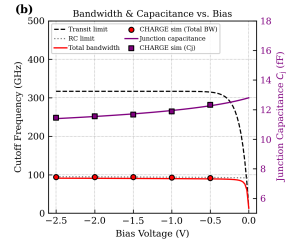
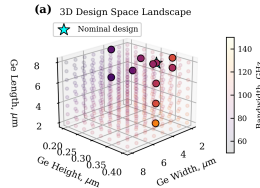
- **SSAC Bandwidth:** Nominal $f_{3dB} = 89.7$ GHz (reaches 93.1 GHz at -1.0 V)
- **Baseline:** $f_{3dB,base} = 56.1$ GHz (no peaking) exceeds the DR4 requirement by 40%
- **Transit-time limit:** $f_{tt} = 66.3$ GHz dominates baseline response prior to inductive peaking



Result 6: Cutoff Frequency Sweeps and Limits

Bandwidth Sweeps

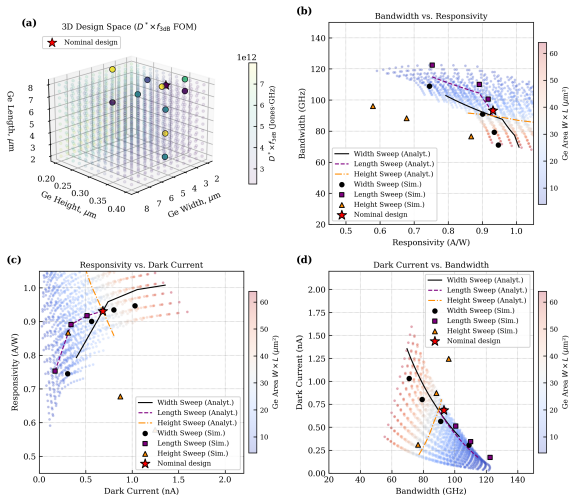
- **3D Landscape:** Cutoff frequency mapped vs. Ge mesa dimensions
- **Bias & Doping:** Bandwidth saturates as junction depletes (> -0.5 V)
- **Screening Roll-off:** Space-charge screening degrades bandwidth above 5 mW



Result 7: Design Trade-offs and Optimization

Design Space Exploration

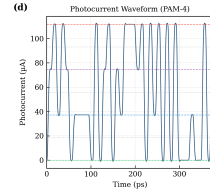
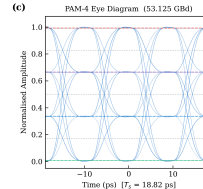
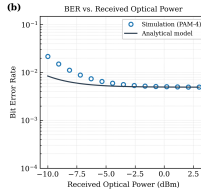
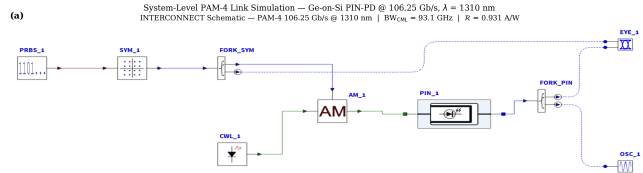
- FoM = $D^* \times f_{3dB}$ [Jones·GHz]: penalises high I_d and low bandwidth simultaneously
- Wider mesa: $\uparrow \mathcal{R}$ (better overlap) but $\uparrow I_d$ (larger SRH volume)
- Nominal $W_m = 5 \mu\text{m}$ achieves $D^* = 3.41 \times 10^{10}$ Jones
- At $W_m = 2 \mu\text{m}$: $D^* = 3.02 \times 10^{10}$ Jones (noise-critical optimum)



Result 8: System-Level PAM-4 Link at 106.25 Gb/s

End-to-End Link Validation

- ▶ 106.25 Gb/s PAM-4 (53.125 GBd) over 500 m SMF at 1310 nm
- ▶ BER: 6.35×10^{-3} at -10 dBm $\rightarrow$ floor
 4.93×10^{-3} at $+3$ dBm (ISI: $Q_{\text{isi}} = 2.48$)
- ▶ OOK sensitivity: $P_{\text{sens}} = 0.955 \mu\text{W}$ (-30.20 dBm) at $Q = 7.03$
- ▶ Clear PAM-4 eye levels with distinct decision thresholds



Nominal Performance Summary

At -1.0 V, $\lambda = 1310$ nm, $T = 300$ K. Benchmark: Shi et al. (2024).

Metric	Source	Nominal Value	Benchmark
Responsivity $\mathcal{R}$	CHARGE solver	0.931 A/W	0.95 A/W
EQE η_{ext}	Coupled chain	88.08 %	91.63 % (FDTD limit)
IQE η_{int}	Carrier collection	96.23 %	Fully depleted junction
Dark current I_d	Electrical solver	0.684 nA	1.3 nA
Junction capacitance C_j	Capacitance solver	11.87 fF	Effective parallel-plate
3dB Bandwidth $f_{3\text{dB}}$	AC solver	89.7 GHz	103 GHz (peaked)
Specific Detectivity D^*	From I_d	3.41×10^{10} Jones	2.95×10^{10} Jones
OOK Sensitivity P_{sens}	Noise model	-30.20 dBm	$Q = 7.03$
PAM-4 Link BER floor	INTERCONNECT	4.93×10^{-3}	$Q_{\text{isi}} = 2.48$

Conclusions and Future Outlook

Key Conclusions

- ▶ $f_{3\text{dB}} = 89.7 \text{ GHz}$, representing $2.25\times$ the DR4 minimum of 39.8 GHz
- ▶ $I_d = 0.684 \text{ nA}$ ($1.9\times$ lower than the experimental benchmark)
- ▶ $\mathcal{R} = 0.931 \text{ A/W}$ within 2 % of reference; EQE 88 % close to FDTD ceiling
- ▶ Full DR4 PAM-4 link validated at 106.25 Gb/s

Future Technical Directions

- ▶ **APD:** model avalanche multiplication for a 10 dB to 15 dB sensitivity gain
- ▶ **Thermal:** simulate self-heating under high CW power
- ▶ **Process corners:** fabrication tolerance sweep to bound yield
- ▶ **TIA co-design:** integrate transimpedance amplifier noise figure

Thank You

Questions & Discussion

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